

1) A college wants to identify students who may fail in the final exam. How can Data Science help?

Data Science analyzes students' attendance, marks, and assignments to predict which students are at risk of failing. Teachers can then provide extra support.

2) A hospital has patient records for many years. How can Data Science be applied?

Data Science analyzes patient records to identify disease patterns, predict illnesses, improve treatments, and support better healthcare decisions.

3) How An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?

Recommendation system - It suggests products based on the customer's browsing history, purchases and preferences.

4. A bank wants to detect suspicious transactions. How can Data Science be useful?

Data Science uses anomaly detection and machine Learning to identify unusual transactions and prevent fraud.

5) A company wants to reduce customer loss. How can Data Science support this?

Data Science predicts customer churn and helps the company take actions such as

personalized offers to retain customers.

7. A school wants to improve student performance. mention one benefit of data science in this case.

It identifies weak students early and helps teachers provide personalized learning support.

8. A supermarket wants to know which products sell more during festivals. How can Data science help?

Data science analyzes past sales data to identify popular products and improve inventory planning.

9. A transport department wants to reduce traffic congestion. How can Data science be applied?

It analyzes traffic patterns and predicts congestion to optimize traffic signals and route planning.

10. A farmer wants to improve crop yield. give one use of Data science.

Data science analyzes soil, weather and crop data to recommend the best farming practices.

11. A company has data in excel tables, customer reviews, images and videos. which facet of data is shown here?

Variety - Different types and formats of data.



11) A weather app updates temperature every 5 minutes. Which facet of data is involved?

Velocity data is generated and updated continuously.

12) A university stores student details in rows and columns. What type of data structure is this?

Structured data.

13) Customer comments on social media are collected for analysis. What type of data is this?

Unstructured data.

14) A company collects a very large amount of daily user activity data from its mobile app. Which Big data characteristic is applied?

Volume - A huge amount of data is collected.

15) A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?

Poor data quality (Accuracy and completeness issues).

16) A business wants to convert customer data into useful sales decisions. Which Big data characteristic is important?

Value - extracting useful insights from data.

17) A company stores large files in Hadoop Distributed file system. Which part of the Big data ecosystem is used?

HDFS (Hadoop Distributed file system)

18) A company wants to process large datasets using parallel computation. which Big Data tool can be used?

~~Apache Spark.~~

19) An organization wants to collect live sensor data from IOT devices. which Big Data ecosystem component is suitable?

~~Apache Kafka (used for real time data streaming).~~

20) A company wants to visualize Big Data results for decision-makers. which tool or method can be applied?

~~Tableau for data visualization.~~

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